RSI(10) / WMA(20) Crossover

on NIFTY and BANKNIFTY index futures

Backtest report — Round 72 (30-minute bars) and Round 73 (daily bars)

Strategy, as specified

- Compute **RSI with length 10** on the bar closes.
- Compute a **weighted moving average (WMA) with length 20** of that RSI series (linear weights 1…20).
- **Long** when the WMA is below the RSI; **short** when the WMA is above the RSI.
- Always in the market, flipping on every cross. 1 lot of the index future.

	Round 72	Round 73
Bar size	30 minutes (lengths counted in bars)	Daily (lengths counted in bars)
Price source	MODELLED PROXY: spot-index bars + modelled carry and rolls	REAL futures: Kite continuous daily series, roll-adjusted
Instruments	NIFTY (primary), BANKNIFTY (secondary)	NIFTY (primary), BANKNIFTY (secondary)
Window	2021-10-01 → 2026-10-09 (5.0 years)	NIFTY 2010-10-14 → 2026-10-13 (16.0 y); BANKNIFTY 2011-03-08 → 2026-10-13 (15.6 y)
Train / test split	≤ 2024-09-30 / 2024-10-01 onward	≤ 2020-12-31 / 2021-01-01 onward
Verdict	REJECT	REJECT

Report date: 29 October 2026 · Prepared for Dheeraj · Source: docs/research/strategy/rsi_wma_crossover/ (R72 2026-10-11, R73 2026-10-14) and strategies/STRATEGY_REGISTRY.md rows 72–73.

1. Executive summary

Verdict: REJECT, on both bar sizes, both indices, in-sample and out-of-sample. The signal has a thin edge before costs. It trades often enough that costs are larger than that edge. On daily bars it trades 12× less, but the edge per trade does not grow to cover the cost, and in recent years it disappears.

The four numbers that carry the result (NIFTY, always-in, lot size in force):

R72 · 30m proxy · 2021-10 → 2026-10	Net **−₹5.44 lakh** over 5 years (gross +₹7.35L, costs ₹12.80L); 536 trades a year
R73 · real daily · same 5-year window	Net **−₹2.05 lakh** at 44 trades a year: gross **−₹354** per trade against **₹572** of cost per trade
R73 · real daily · full 16 years	Net **+₹10,800** (t = 0.02, i.e. breakeven). Excluding 2020 alone: **−₹3.33 lakh**
R73 · test period 2021 → 2026	**−₹3.16 lakh**. Every variant negative; 9 of 9 nearby RSI/WMA settings negative, also at zero slippage

- **The two rounds are different kinds of evidence.** R72 is a modelled proxy, because no intraday futures history exists for expired contracts. R73 uses real futures prices. Read §3 before comparing their rupee figures.
- **Turnover is the failure.** On 30-minute bars the signal flips about 540 times a year. Each round trip costs more than it earns on average, and the gap widens as STT rises (0.01% → 0.05% since 2021).
- **Daily bars cut turnover, not the gap.** Cost per trade rises, because 22% of 5-day holds pay a roll. Gross per trade on the recent window turns negative. See the turnover exhibit in §5.
- **One year carries the 16-year daily result.** 2020 (crash then rebound) made +₹3.44L on NIFTY. Every other year together is −₹3.33L.
- **A 1-lot account does not survive it.** One NIFTY lot needs about ₹1.65L of margin, but staying solvent through the drawdowns needed ₹2.9–5.8L depending on the start date. At the account's current cash level, the first margin call would have come within months of a 2021 start.
- **Recommendation:** do not deploy, live or sandbox. Long-only is mostly index beta, which buy-and-hold carries without ~22 round trips a year.

2. Methodology and data provenance

Do not mix the two rounds up. R72 is a modelled proxy: the signal runs on spot-index 30m bars, and futures P&L is built on top with a modelled basis, carry and rolls. **R73 is real futures:** Kite's continuous daily futures series, stitched month to month, with carry already inside the prices. R73's numbers are the stronger evidence; R72's are the best available estimate for an intraday version.

2.1 Data

Item	R72 (30-minute)	R73 (daily)
Signal prices	NSE spot index 30m bars from Kite (NIFTY 50, NIFTY BANK), 2021-06 → 2026-10; warm-up before 2021-10-01	Futures daily closes from Kite **continuous=1** on the live front-month token
P&L prices	Modelled futures: spot × (1 + c × days-to-expiry / 365)	Real stitched futures, difference back-adjusted at each roll
History available	Kite serves intraday bars only for contracts still trading, so 5 years of real 30m futures does not exist	NIFTY from 2010-10-14 (3,966 days); BANKNIFTY from 2011-03-08 (3,867 days)
Cross-check	Signal on the real Oct-2026 contract (2.5 months): NIFTY 2.9 vs 2.4 points/flip on spot	Kite spot daily vs historify daily from 2022: 1,166 of 1,177 days match within 0.01%

2.2 Execution, variants and split

- Signal evaluated at each bar's close; the trade executes at the **next bar's open** (no look-ahead). On 30m bars a 15:15 signal executes at the next day's 09:15 open, so overnight gaps are included.
- Variants (identical across rounds): **always_in** (as specified); **flat_between** (long only if RSI is also above 50, short only if below 50, otherwise flat); **long_only**. R72 adds **intraday_flat** (always-in but flat at every close).
- Two lot conventions: the **lot size in force** for each contract, and a **fixed** 65 (NIFTY) / 30 (BANKNIFTY), today's lots. No parameter was fitted, so the train/test split asks one question: does the result hold in a later regime?

2.3 Carry and roll handling

- **R72 carry** was calibrated from 1,157 matched 30m bars of the Oct/Nov/Dec-2026 contracts vs the index: **NIFTY 6.32%/yr** (middle-50% range 5.69–6.82%), **BANKNIFTY 7.38%/yr**. Results were re-run at ×0.5, ×1.5 and ×0. Rolls: a round trip at the 15:15 bar of each monthly expiry held through (last Thursday to Aug-2025, last Tuesday from Sep-2025, holiday-adjusted).
- **R73 carry** is inside the real prices. Kite's raw stitch jumps by the new contract's premium at each roll (median +0.40% NIFTY, +0.44% BANKNIFTY), which would count as fake profit for longs. The series is back-adjusted using the new contract's observed basis applied to the prior day's spot close. Check: the adjusted series runs 5.2%/yr (NIFTY) and 5.8%/yr (BANKNIFTY) below spot buy-and-hold, close to R72's calibration. Roll days are detected from the data (largest basis jump in each month's last 10 days); a calendar-roll re-run moved NIFTY by about ₹6k.

2.4 Cost model (per leg, at the schedule in force on each trade date)

Component	Rate	Notes
Brokerage	min(₹20, 0.03% of turnover)	Zerodha F&O; a flip is costed as two legs
STT (sell side only)	0.017% → 0.01% (2013-06-01) → 0.0125% (2023-04-01) → 0.02% (2024-10-01) → **0.05% (2026-04-01)**	Budget 2026 raised futures STT 2.5×
Exchange transaction	0.0019%; 0.00173% from 2024-10-01	NSE futures
SEBI turnover fee	₹10 per crore	
Stamp duty (buy side)	0.002%	
GST	18% on brokerage + exchange + SEBI	
Slippage	R72: NIFTY 1.0 pt, BANKNIFTY 2.5 pt per leg. R73: 0.004% / 0.0045% of price per leg (the same at today's levels)	Zero slippage shown as the upper bound; 2× as a stress
Roll	Two extra legs at each expiry held through	Same cost schedule

2.5 Lot sizes in force

Index	Lot size by contract expiry (verified against NSE circulars / broker notices)
NIFTY	25 (to Oct-2015) → 75 (Nov-2015 → Jun-2021) → 50 (Jul-2021 → Apr-2024) → 25 (May-2024 → Jan-2025) → 75 (Feb-2025 → Dec-2025) → 65 (from Jan-2026)
BANKNIFTY	Verified from Nov-2018 only: 20 → 25 (Jul-2020) → 15 (Jul-2023) → 30 (Feb-2025) → 35 (Jul-2025) → 30 (Jan-2026). Pre-2018 sizes are unverified, so R73 BANKNIFTY at the lot in force starts 2018-11; the full 16 years use the fixed 30 lot.

3. Results — Round 72 (30-minute bars, modelled proxy)

3.1 NIFTY always-in (as specified)

Lot	Period	Trades	Win %	Gross ₹	Costs ₹	Net ₹	Payoff	Max DD ₹	Sharpe	t (est.)
in force	**all**	**2,690**	**29.5**	**+7,35,444**	**12,79,688**	**−5,44,244**	**2.17**	**−7,71,585**	**−0.64**	**−1.43**
in force	train	1,579	30.1	+3,39,999	4,78,033	−1,38,034	2.20	−3,29,278	−0.37	−0.64
in force	test	1,111	28.6	+3,95,444	8,01,655	−4,06,210	2.18	−5,54,985	−0.91	−1.29
fixed 65	all	2,690	29.6	+10,08,622	14,66,553	−4,57,931	2.22	−7,43,015	−0.46	−1.03
fixed 65	train	1,579	30.3	+4,26,109	6,52,241	−2,26,132	2.16	−4,99,804	−0.42	−0.73
fixed 65	test	1,111	28.6	+5,82,513	8,14,312	−2,31,799	2.32	−4,90,505	−0.52	−0.74

Trades per year ≈ 536 (hold ≈ 3 hours). Gross per trade +₹273, cost per trade ₹476 (in force, all); in the test period +₹356 vs ₹722. R72 did not compute per-trade t-statistics; the t column is approximated as Sharpe × √years from the daily P&L series.

3.2 Long vs short (NIFTY always-in, lot in force)

Period	Long trades	Long win %	Long net ₹	Short trades	Short win %	Short net ₹
all	1,345	28.8	−3,28,804	1,345	30.1	−2,15,440
train	789	30.5	+15,985	790	29.6	−1,54,019
test	556	26.4	−3,44,789	555	30.8	−61,421

3.3 Variants side by side (lot in force)

Index	Variant	Trades	Gross	Net	Sharpe	Train net	Test net	Max DD
NIFTY	**always_in (spec)**	**2,690**	**+7.35L**	**−5.44L**	**−0.64**	**−1.38L**	**−4.06L**	**−7.72L**
NIFTY	flat_between	2,209	+8.70L	−1.72L	−0.25	+0.69L	−2.41L	−4.22L
NIFTY	long_only	1,345	+3.11L	−3.29L	−0.51	+0.16L	−3.45L	−4.03L
NIFTY	intraday_flat	3,789	+2.76L	−15.04L	−2.35	−5.27L	−9.77L	−16.30L
BANKNIFTY	**always_in (spec)**	**2,724**	**+11.28L**	**−2.24L**	**−0.22**	**−0.07L**	**−2.18L**	**−5.47L**
BANKNIFTY	flat_between	2,277	+10.25L	−1.00L	−0.12	+1.48L	−2.49L	−3.94L
BANKNIFTY	long_only	1,362	+5.71L	−1.05L	−0.15	+0.57L	−1.62L	−2.31L
BANKNIFTY	intraday_flat	3,812	+6.10L	−12.61L	−1.64	−3.48L	−9.13L	−14.12L

3.4 Sensitivity (lot in force; net, all period [train / test])

Case	NIFTY always_in	NIFTY flat_between	NIFTY long_only	BANKNIFTY always_in
base	−5.44L [−1.38 / −4.06]	−1.72L [+0.69 / −2.41]	−3.29L [+0.16 / −3.45]	−2.24L [−0.07 / −2.18]
slippage 0 (upper bound)	−2.52L [+0.10 / −2.62]	+0.67L [+1.91 / −1.25]	−1.83L [+0.90 / −2.73]	+1.09L [+1.65 / −0.55]
slippage ×2	−8.36L	−4.11L	−4.75L	−5.58L
carry ×0.5 / ×1.5	−5.41L / −5.47L	−1.68L / −1.76L	−2.37L / −4.21L	−2.27L / −2.22L
today's costs throughout	−14.73L	−9.31L	−7.93L	−11.63L

3.5 Per year (NIFTY always-in, lot in force)

Year	Trades	Win %	Gross ₹	Costs ₹	Net ₹	Long net ₹	Short net ₹
2021 (Oct–Dec)	127	38.6	+1,36,132	38,305	+97,828	+40,348	+57,480
2022	506	33.2	+1,26,677	1,50,323	−23,646	−20,547	−3,099
2023	546	29.1	+20,002	1,79,679	−1,59,677	−20,310	−1,39,367
2024	535	26.4	+1,23,405	1,45,347	−21,943	+113	−22,055
2025	571	27.7	−58,062	3,81,054	−4,39,116	−1,70,106	−2,69,011
2026 (to Oct-09)	405	29.1	+3,87,290	3,84,979	+2,310	−1,58,302	+1,60,612

3.6 Nearby settings (always-in, fixed lot, net ₹)

RSI / WMA	NIFTY train	NIFTY test	NIFTY test, slip 0	BANKNIFTY train	BANKNIFTY test	BANKNIFTY test, slip 0
7 / 10	−5,30,361	−8,40,454	−6,40,709	−2,74,740	−6,81,830	−4,36,805
7 / 20	−1,93,515	−4,63,528	−3,01,873	+17,472	−2,36,449	−49,174
7 / 30	−1,32,766	−3,26,484	−1,82,769	+58,407	−1,43,461	+24,014
10 / 10	−5,39,150	−7,52,771	−5,61,866	−1,52,865	−4,72,905	−2,40,030
10 / 20 (spec)	**−2,26,132**	**−2,31,799**	**−84,834**	**−17,988**	**−49,644**	**+1,18,431**
10 / 30	+1,60,445	−1,81,662	−57,707	+1,62,045	−1,53,499	−8,824
14 / 10	−5,12,858	−5,93,240	−4,10,395	−62,339	−4,82,315	−2,60,240
14 / 20	−92,328	−1,40,385	−8,500	+90,946	−51,905	+1,00,570
14 / 30	+38,070	−87,371	+23,324	−1,00,839	−85,775	+50,350

All 18 test cells are negative at the stated slippage. The best train cell (10/30) loses in test on both indices.

4. Results — Round 73 (daily bars, real futures)

4.1 NIFTY, 2010-10-14 → 2026-10-13

Variant / lot	Period	Trades	Tr/yr	Win %	Gross ₹	Costs ₹	Net ₹	Payoff	Max DD ₹	Sharpe	t
always_in · in force	**all**	**688**	**43.0**	**35.2**	**+2,40,182**	**2,29,382**	**+10,800**	**1.85**	**−5,69,864**	**0.01**	**0.02**
	train	428	41.9	36.9	+4,14,939	88,393	+3,26,546	2.08	−1,96,463	0.29	0.81
	test	260	45.0	32.3	−1,74,757	1,40,989	−3,15,746	1.79	−4,88,031	−0.29	−0.84
always_in · fixed 65	all	688	43.0	35.5	+2,08,441	2,57,417	−48,976	1.80	−6,46,426	−0.02	−0.08
	train	428	41.9	37.1	+4,61,508	99,539	+3,61,969	2.03	−1,70,457	0.33	0.96
	test	260	45.0	32.7	−2,53,068	1,57,878	−4,10,945	1.73	−6,41,956	−0.35	−0.93
flat_between · in force	all	591	37.0	35.5	+1,58,891	1,88,571	−29,680	1.80	−4,37,830	−0.02	−0.06
	train / test	374 / 217					+2,79,677 / −3,09,358			0.29 / −0.34	0.78 / −0.94
long_only · in force	all	344	21.5	36.6	+3,29,553	1,09,867	+2,19,686	1.95	−3,22,052	0.17	0.66
	train / test	214 / 130					+3,30,209 / −1,10,523			0.46 / −0.13	1.68 / −0.41

Gross per trade / cost per trade (always-in, in force): all +₹349 / ₹333; train +₹969 / ₹207; test −₹672 / ₹542. Average hold 5.7 trading days. t = mean net per trade ÷ standard error.

4.2 BANKNIFTY

Variant / lot	Window	Trades	Net ₹	Train net ₹	Test net ₹	Sharpe	t (all)
always_in · fixed 30	2011-03 → 2026-10	692	+75,505	+2,62,891	−1,87,386	0.03	0.10
always_in · in force	2018-11 → 2026-10	349	+1,10,927	+2,19,201	−1,08,273	0.07	—
flat_between · fixed 30	2011-03 → 2026-10	586	−91,672	+4,00,924	−4,92,595	−0.04	−0.14
long_only · fixed 30	2011-03 → 2026-10	346	+3,09,033	+2,90,143	+18,890	0.16	0.65

BANKNIFTY always-in fixed: gross per trade +₹510, cost per trade ₹401, max DD −₹7.73L. Without 2020 (+₹5.23L) the 16-year net is −₹4.47L. Long-only's +₹18,890 test result is t = 0.05 and turns to −₹44k at today's STT.

4.3 Long vs short (NIFTY always-in, in force)

Period	Long trades	Long win %	Long net ₹	Short trades	Short win %	Short net ₹
all (16 y)	344	36.6	+2,19,686	344	33.7	−2,08,886
train	214	40.2	+3,30,209	214	33.6	−3,662
test	130	30.8	−1,10,523	130	33.8	−2,05,224

The long side's 16-year gain is largely the index's upward drift; the short side pays it back. Both sides lose in the test period.

4.4 Sensitivity (fixed lot; net, all period [train / test])

Case	NIFTY always_in	NIFTY long_only	BANKNIFTY always_in	BANKNIFTY long_only
base	−49k [+3.62L / −4.11L]	+1.91L [+3.36L / −1.45L]	+76k [+2.63L / −1.87L]	+3.09L [+2.90L / +19k]
slippage 0	+8k [+3.85L / −3.77L]	+2.19L [/ −1.29L]	+1.42L [/ −1.48L]	+3.42L [/ +38k]
slippage ×2	−1.06L	+1.63L	+9k	+2.76L
today's costs throughout	−2.91L [/ −5.43L]	+73k [/ −2.08L]	−1.72L [/ −3.18L]	+1.88L [/ −44k]
calendar rolls	−43k [/ −4.10L]	+2.36L [/ −1.38L]	+1.20L [/ −1.62L]	+4.43L [/ +87k]

4.5 Per year — NIFTY always-in (lot in force) and BANKNIFTY always-in (fixed 30), net ₹

Year	Trades	Win %	Gross ₹	Costs ₹	NIFTY net ₹	Long net	Short net	BANKNIFTY net ₹
2010 (Oct–Dec)	4	25.0	−6,798	471	−7,269	−473	−6,796	—
2011	45	37.8	+24,411	4,920	+19,491	−9,988	+29,479	+22,065
2012	41	39.0	+4,739	4,567	+172	+11,435	−11,263	+40,450
2013	42	40.5	+5,735	4,752	+982	−1,721	+2,703	−91,754
2014	38	26.3	+8,992	4,461	+4,531	+21,043	−16,512	+37,076
2015	36	52.8	+55,479	5,324	+50,155	+6,255	+43,900	−74,196
2016	56	32.1	−21,188	12,646	−33,834	−10,663	−23,171	+15,425
2017	36	50.0	+21,841	10,794	+11,046	+75,109	−64,063	+12,586
2018	44	31.8	+38,958	13,085	+25,873	+12,390	+13,483	+11,379
2019	42	28.6	−75,215	13,507	−88,722	−12,254	−76,468	−2,32,697
2020	**44**	**36.4**	**+3,57,985**	**13,864**	**+3,44,121**	**+2,39,075**	**+1,05,045**	**+5,22,557**
2021	48	29.2	−85,692	17,102	−1,02,793	+39,763	−1,42,557	−81,330
2022	38	34.2	+91,863	12,195	+79,668	+25,375	+54,293	+3,48,181
2023	47	29.8	+35,955	16,241	+19,714	+75,452	−55,739	+1,98,559
2024	40	30.0	−1,41,354	13,662	−1,55,017	−63,955	−91,062	−4,65,465
2025	46	37.0	−54,488	36,890	−91,378	−6,214	−85,163	−1,09,977
2026 (to Oct-13)	41	34.1	−21,042	44,899	−65,940	−1,80,945	+1,15,004	−77,354

The highlighted 2020 row is the whole 16-year result. NIFTY 2010–15 rupee figures are small because the lot was 25. Costs per year rose from about ₹5k (2011–15) to about ₹45k (2026): the lot tripled and STT went from 0.01% to 0.05%.

4.6 Nearby settings on the TEST period (always-in, fixed lot, net ₹)

RSI / WMA	NIFTY train	NIFTY test	NIFTY test, slip 0	BANKNIFTY test	BANKNIFTY test, slip 0
7 / 10	−59,768	−4,36,817	−3,93,114	−3,00,185	−2,50,118
7 / 20	+2,45,998	−2,87,785	−2,49,660	−41,582	−189
7 / 30	+4,66,063	−4,45,453	−4,08,995	−1,02,490	+24,014
10 / 10	+94,968	−4,01,705	−3,60,093	+4,334	+52,600
10 / 20 (spec)	**+3,61,969**	**−4,10,945**	**−3,76,762**	**−1,87,386**	**−1,47,736**
10 / 30	+5,65,898	−2,57,647	−2,26,322	−1,11,988	−76,720
14 / 10	+2,64,464	−3,13,912	−2,72,207	−19,920	+27,033
14 / 20	+5,32,095	−3,78,860	−3,46,285	−2,25,613	−1,87,135
14 / 30	+6,09,178	−2,86,102	−2,59,161	−1,53,095	−1,19,732

NIFTY: 8 of 9 cells positive in train, 9 of 9 negative in test (also at zero slippage). The best train settings are not the ones that hold up — the in-sample pattern earlier rounds (R61/R62) warned about. BANKNIFTY: 1 of 9 positive in test (+₹4k, noise).

5. The turnover comparison — the central exhibit

The question R73 was run to answer: if daily bars lengthen the holding period, does gross profit per trade rise enough to cover cost per trade? NIFTY always-in, lot size in force.

	Bars	Window	Trades / yr	Hold	Gross / trade	Cost / trade	Gap / trade	Net
R72 (proxy)	**30m**	**2021-10 → 2026-10**	**536**	**≈ 3 h**	**+₹273**	**₹476**	**−₹203**	**−₹5.44L**
R73 (real)	**daily**	**same window**	**44**	**5.6 d**	**−₹354**	**₹572**	**−₹926**	**−₹2.05L**
R73 (real)	daily	full 16 y	43	5.7 d	+₹349	₹333	+₹16	+₹0.11L
R73 (real)	daily	train 2010–20	42	5.9 d	+₹969	₹207	+₹762	+₹3.27L
R73 (real)	daily	test 2021–26	45	5.5 d	−₹672	₹542	−₹1,214	−₹3.16L

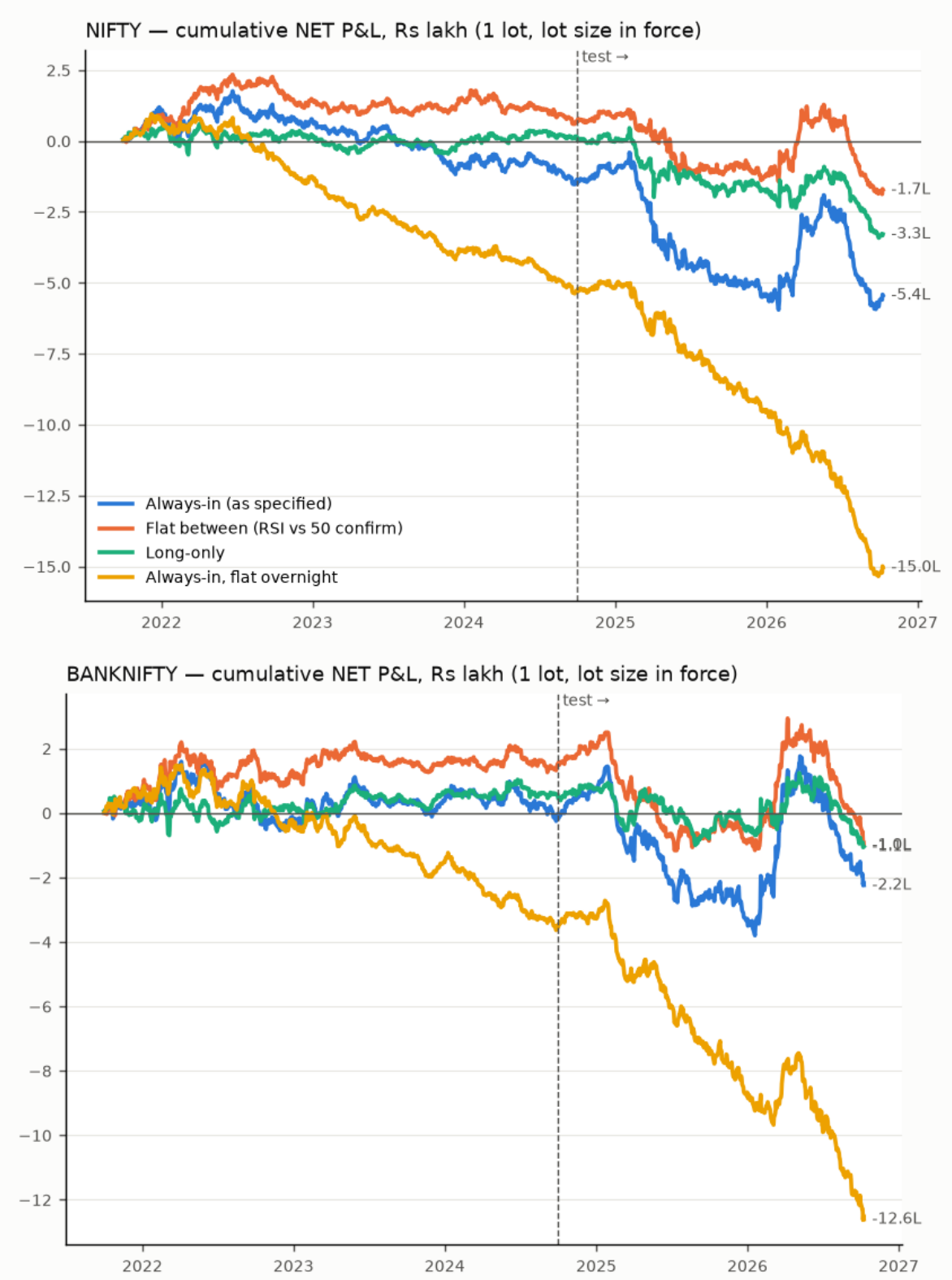
Answer: the gap does not close.

- **Trades per year fell 12×** (536 → 44), as expected.
- **Cost per trade rose** to ₹572 instead of falling: a 5–6 day hold crosses an expiry on 22–24% of trades and pays a roll round trip, on the same lot notional.
- **Gross per trade fell** to −₹354 on the same window, i.e. the daily signal had no edge at all in 2021–26.
- **Over 16 years gross ≈ cost** (+₹349 vs ₹333): positive 2010–20, negative 2021–26.
- The daily version loses less than the 30m one only because it trades less.

Note on comparability: R72's figures come from modelled futures (spot signal plus modelled carry), R73's from real futures. The same-window rows are the closest like-for-like comparison available; a short overlap check (§2.1) suggests the R72 proxy does not understate real futures results.

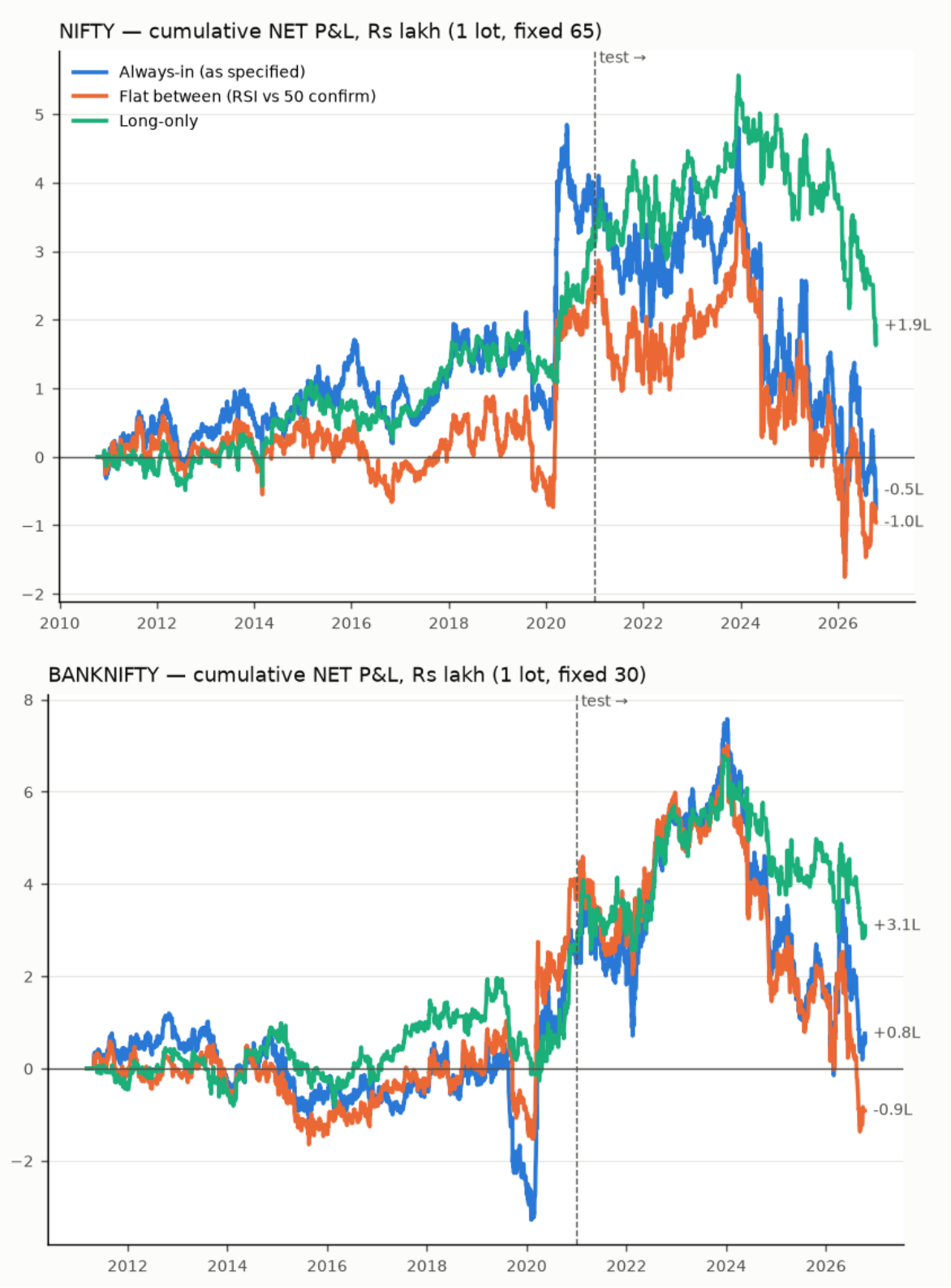
6. Equity curves

6.1 R72 — 30-minute bars, modelled proxy (cumulative net P&L, ₹ lakh, lot size in force)



Variants: always-in (blue), flat-between (orange), long-only (green), always-in flat overnight (amber). Dashed line = start of the test period (2024-10-01). MODELLED PROXY: spot-index 30m signal + modelled futures P&L (carry, roll, full Zerodha costs, slippage NIFTY 1 pt / BANKNIFTY 2.5 pt per leg). Not real futures fills.

6.2 R73 — daily bars, real continuous futures (cumulative net P&L, ₹ lakh, fixed lot)



Variants: always-in (blue), flat-between (orange), long-only (green). Dashed line = start of the test period (2021-01-01). The 2020 step is visible on both indices. REAL futures prices: Kite continuous daily futures, roll-adjusted; full Zerodha costs at the STT in force; slippage about 0.004% of price per leg; roll round trip at every expiry held.

7. Survivability — can a 1-lot account carry it?

Margin for one lot today (Kite margin calculator, NRML, 2026-10-09): **NIFTY ₹1,64,713** (SPAN ₹1,35,301 + exposure ₹29,413, on about ₹14.7L notional) and **BANKNIFTY ₹1,86,441** (about ₹16.7L notional). Historic margins are estimated at today's ratio, 11.2% of notional.

Case	Start	Max drawdown	Capital never to be margin-called*	First margin call at current cash
R72 · NIFTY always-in · in force	2021-10 start	−₹7.72L	4.7× margin	Liquidated in the 2025 leg (−₹4.4L in one year)
R73 · NIFTY always-in · in force	2010-10 start	−₹5.70L	~₹2.88L	Feb-2026
R73 · NIFTY always-in · in force	2021-01 start		~₹5.81L	May-2021
R73 · NIFTY always-in · in force	2021-10 start		~₹5.33L	Jan-2025
R73 · NIFTY always-in · fixed 65	2010-10 / 2021-01 / 2021-10	−₹6.46L	~₹3.48L / ₹6.80L / ₹6.42L	— / Jul-2021 / Dec-2021
R73 · BANKNIFTY always-in · in force	2018-11 start	−₹5.32L	~₹2.74L	—
R73 · BANKNIFTY always-in · fixed 30	2011-03 start	−₹7.73L	~₹4.32L	—

** Starting capital such that capital plus cumulative P&L always covers the open position's margin (11.2% of futures notional). "Current cash" is the account's available cash at the 2026-10-09 check, which covers one NIFTY lot with under ₹20k of headroom and does not cover one BANKNIFTY lot; the exact figure is deliberately left out of this report. — = not computed.*

- A 2010 start looks better only because it banks the 2020 windfall before the losses arrive; the binding point is Feb-2026, after the 2024–26 losses erased it.
- From the start of the test period (2021) the strategy needed 3.2–4.1× one lot's margin to stay solvent.
- R73 survives better than R72 (smaller drawdown), but at today's cash neither survives.

8. Limitations and caveats

- **R72 is modelled.** Real 30m futures for expired contracts are not available from Kite. The overlap check with the live Oct-2026 contract covers only about 120 flips, too few to calibrate a spot-to-futures adjustment.
- **Sample size.** R73 has 688 NIFTY trades over 16 years but only 260 in the test period. That is enough to see the train-to-test sign flip on both indices and every variant, but not to size the test loss precisely. No variant reaches t = 2 in any period.
- **Single-source futures series.** R73's futures prices come only from Kite's continuous series; historify holds no futures. The spot series was cross-checked (1,166/1,177 days), the futures series was not. Roll dates are inferred from basis jumps; using the expiry calendar instead changed NIFTY by about ₹6k and BANKNIFTY by up to ₹1.3L, with no change of sign that matters.
- **BANKNIFTY lot sizes before Nov-2018** could not be verified, so BANKNIFTY's lot-in-force results start in 2018-11; the full-history BANKNIFTY figures use the fixed 30 lot.
- **2020 carries everything.** The 16-year daily result depends entirely on one crash-and-rebound year. Without it NIFTY is −₹3.33L and BANKNIFTY −₹4.47L.
- **STT regime change.** Futures STT moved from 0.01% to 0.05% between 2023 and 2026. Results at today's schedule are worse in every case; history run on old rates overstates what the strategy would earn now.
- **Margin is estimated** for past years at today's SPAN ratio; actual historical SPAN varied with volatility.

9. Conclusion and what would be worth testing next

Conclusion. The RSI(10)/WMA(20) crossover on index futures does not have a tradeable edge. On 30-minute bars it earns a thin gross edge and pays more than that in costs. On daily bars, using real futures over 16 years, it is breakeven, all of it from one year, and it loses in every variant and every nearby setting in 2021–26. It also cannot be carried on a one-lot account. Not recommended for live or sandbox.

Nothing here argues for more work on this exact signal. If the idea is pursued, these are the only directions the evidence leaves open, each to be pre-registered with an out-of-sample test before any result is trusted:

- **Much lower turnover.** The one consistent pattern is that slower settings lose less because they trade less. A weekly-bar version, or entries only on RSI crossing 50 with a minimum holding period, would test whether any of the gross edge survives at a few trades a year.
- **Volatility-regime conditioning.** The only money was made in 2020's high-volatility trend. A rule that trades only above a fixed India VIX level is testable on the same real daily futures, but must be fixed in advance; chosen after looking at 2020 it would simply be fitting that year.
- **Real intraday futures data** from a paid vendor would turn R72 from a proxy into a real test. Given that R72 already loses at zero slippage under today's costs, this is worth buying only if a lower-turnover intraday variant is on the table.

Source documents: docs/research/strategy/rsi_wma_crossover/2026-10-11_r72_rsi10_wma20_30m_index_futures.md and 2026-10-14_r73_rsi10_wma20_daily_index_futures.md; registry rows 72–73 in strategies/STRATEGY_REGISTRY.md; GitHub issues #761 and #768.